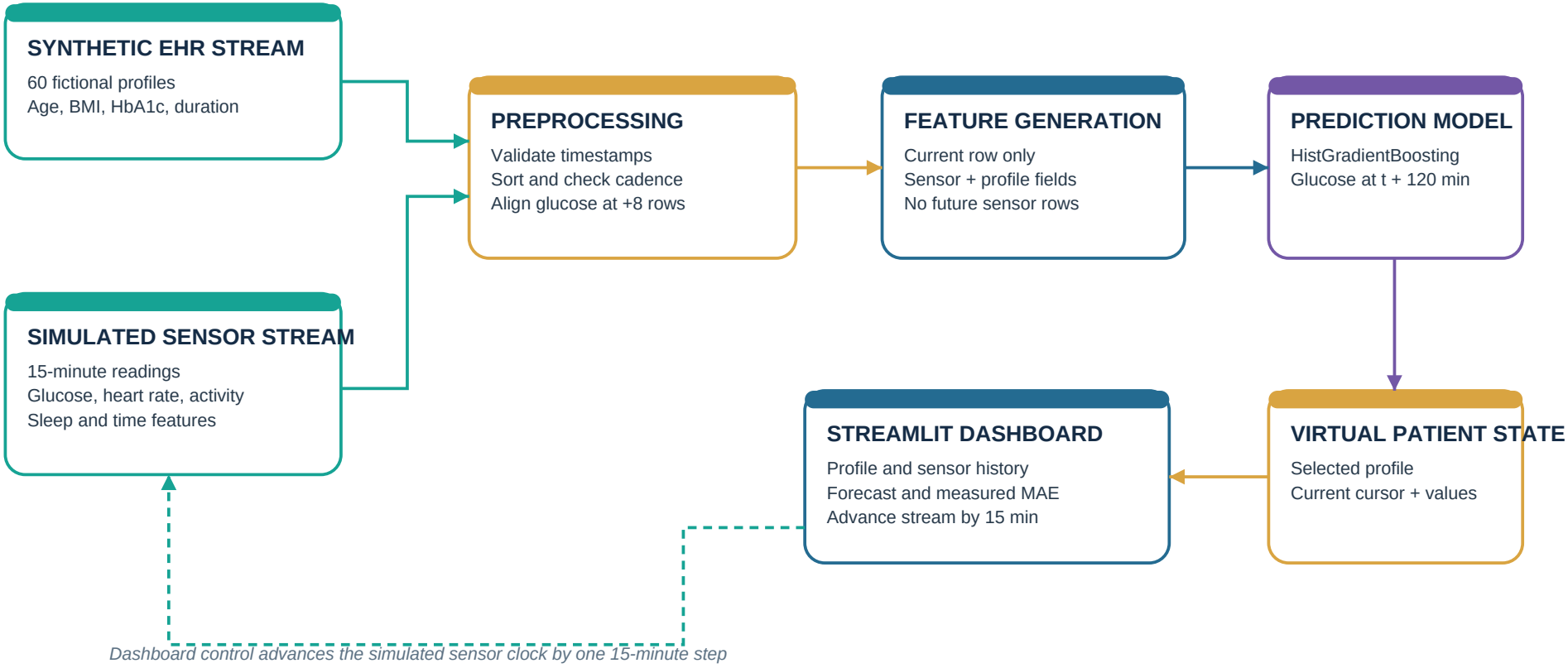
GlucoTwin | System Architecture

Local synthetic two-hour glucose forecasting proof of concept



Measured run: HistGradientBoostingRegressor | train 95,880 | test 24,120 | model MAE 3.335 mg/dL | persistence MAE 8.580 mg/dL

Leakage control: training labels are purged so every training target timestamp precedes the test cutoff. At inference, the model receives only the selected patient profile and the current simulated sensor row. No real patient data or clinical validation are included.